(a) Processing chain

one C-arm sweep up the spine, lateral

shaded = learned; everything else is geometry

expose	detect	link	chain	count	confirm
overlapping lateral exposures up the spine, any device	vertebral bodies and the anchor (sacrum, or C2)	NCC to the previous frame, inside the protocol's window	one node per body, end to end; one panorama	integer traversal from the anchor: an ordinal per node	instrument frame matched onto the chain: its level

the count is an integer traversal of the chain from the anchor, not a per-frame choice among level names

(b) The algorithm on one whole spine, in two states

row (b) is not 13.0 x 2.6 in

fig1_row2_stitch.png: 1997 x 378 px (6.66 x 1.26 in), aspect 5.283 -- needs 5.000 (13.0 x 2.6 in)

(c) Matched crops across the link, with the alias margin

row (c) is not 13.0 x 2.6 in

fig1_row3_confirm.png: 1713 x 509 px (5.71 x 1.70 in), aspect 3.365 -- needs 5.000 (13.0 x 2.6 in)

(d) What is in the data, counted

TEACHES ANATOMY 138,838 labelled vertebrae, film and render

PLAIN FILM ONLY



88,416 labelled vertebrae

VinDr 36,622 · CSXA 21,095

BUU 15,310 · FluoroCap 12,648

NHANES II 2,741

5 corpora · no CT and no pose

dataset_profile.json

SYNTHETIC



50,422 labelled vertebrae

CT renders 31,628 · VerSe 3,837

variants 6,258 · CTSP1K 6,091

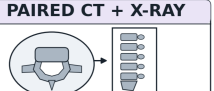
NHANES 2,608 · 4,551 files on disk

CT → render → fitted device card

dataset_profile + RSNAI_DATA_MANIFEST

TEACHES THE DEVICE 4 device cards fitted on 40, 40, 40, 14 frames; held-out R^2 0.68–0.97

PAIRED CT + X-RAY



3,794 frames · 294 subjects

C-ArmLSpine 296 fr, 148 cases

COBRA2026 3,480 fr, 145 cases

Ljubljana 18 fr, 1 cadaver

the same subject in both, pose known

or_ready_audit_track + devices/*.json

C-ARM, NON-SPINE ANATOMY



482 frames · 7 subjects

DeepFluoro 366 fr, 6 cadaveric pelves

Geneva hip 116 fr, 1 resin phantom

the device's settings and its image formation only — never a vertebral level

or_ready_audit_track_2026-09-22.json

PHANTOM



SpineSAC · 6 C-arms

one sawbone spine, imaged raw

5,490 files, 4.1 GB on disk

phantom CT and poses shipped, no bone truth: nothing is scored

RSNAI_DATA_MANIFEST.json

HELD OUT, SCORED NOT FITTED C-ArmLSpine 28 cases · DeepFluoro specimen 18-1109, 104 of 366 frames · the Surgio B51 and SpineSAC's C-arms have no fitted card at all

(e) Training corpus and the two networks

row (e) is not 13.0 x 2.6 in

fig1_row5_corpus_arch.png: 3900 x 1080 px (13.00 x 3.60 in), aspect 3.611 -- needs 5.000 (13.0 x 2.6 in)

(f) Fluoroscopic localisation, with the software's output

row (f) is not 13.0 x 2.6 in

fig1_row6_confirm_fluoro.png: 1857 x 509 px (6.19 x 1.70 in), aspect 3.648 -- needs 5.000 (13.0 x 2.6 in)